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MILITARY SPECIFICATION

HOSE ASSEMBLY, HIGH PRESSURE, BREATHING OXYGEN

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers a high-pressure oxygen hose assembly having a normal working pressure of 2,500 psi.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitations for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-B-621	Boxes, Wood, Nailed and Lock-Corner
NN-B-631	Boxes; Wood, Wirebound, (For Domestic Shipment)
NN-P-515	Plywood, Container Grade
ZZ-R-601	Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analyses)
LLL-B-631	Boxes; Fiber, Corrugated (For Domestic Shipment)
LLL-B-636	Boxes; Fiber, Solid, (For Domestic Shipment)
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material
MIL-N-6011	Nitrogen; Liquid and Gas
MIL-O-8069	Oxygen, Liquid and Gas
MIL-R-6125	Regulator, Oxygen Pressure, Type C-1
MIL-R-8202	Regulator, Diluter Demand, Oxygen, Automatic, Pressure Breathing, Type D-2
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant

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STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

U. S. Air Force

44B27648	Hose Assembly - Oxygen, High Pressure
48B3659	Hose Assembly - Oxygen, High Pressure, Wire-Braided

Bureau of Aeronautics

54A96	Hose Assembly, Oxygen, High Pressure

PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

No. 143	Specifications and Standards; Use of
No. 298	Code Symbols for Hose Manufacturers
No. 405	Storage Life - Aeronautical Articles

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The hose assembly furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Materials.-

3.2.1 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143.

3.3 Design and construction.- The hose assembly shall be designed and constructed in accordance with Drawing 44B27648, 48B3659, or 54A96, as specified by the procuring activity. (See paragraph 6.2.)

3.3.1 The hose shall be fabricated of a compounded inner tube reinforced with braids of wire, a suitable binder over the outer wire braid, and a braided protective cover.

3.3.1.1 Inner tube.- The inner tube shall consist of a nonblooming compound of seamless construction, and shall be as free of pitting as is consistent with good manufacturing practice. The inner tube shall be flushed with water to insure removal of all dirt, foreign material, and excess mandrel lubricant. Any compound that is easily deteriorated, or otherwise adversely affected by continued use with oxygen, shall not be used. The inner tube shall emit no objectionable odor.

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3.3.1.2 Reinforcing material.- The reinforcing shall consist of two wire braids of sufficient strength to meet the requirements specified herein.

3.3.1.3 Outer surface.- The outer surface shall resist abrasion and sunlight deterioration, shall have sufficient porosity to prevent ballooning, and shall be composed or treated to inhibit the growth of fungus.

3.3.2 Physical properties.- The inner tubes shall have the following physical characteristics.

3.3.2.1 Tube tensile strength.- The tensile strength of the inner tube material shall be not less than 1,200 psi.

3.3.2.2 Ultimate elongation.- The ultimate elongation of the inner tube material shall be not less than 200 percent (2 inches extended to 6 inches or more).

3.3.2.3 Accelerated aging.- The inner tube shall be capable of withstanding exposure to oxygen under a pressure of 200 psi at ±160°F.

3.4 Performance.-

3.4.1 High and low-temperature performance.- The hose assembly shall perform satisfactorily at temperatures between -65° and +160°F, and shall show no signs whatsoever of failure.

3.4.2 Hose assembly tensile strength.- The hose assembly shall be able to withstand a tensile pull of 1,000 pounds.

3.4.3 Test pressure.- The hose assembly shall be able to withstand a test pressure of 2,500 psi without leakage.

3.4.4 Burst pressure.- The hose assembly shall not burst, leak, or blow off fittings at pressures less than 15,000-psi hydraulic pressure.

3.5 Dimensions.- The inside diameter of the hose shall be 1/4 ±0.015 inch. The outside diameter shall be 5/8 ±0.020 inch.

3.6 Plugs.- The ends of each hose assembly shall be sealed with suitable metal or plastic plugs to prevent the entrance of dirt or moisture.

3.7 Identification of product.-

3.7.1 Hose.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.7.2 Hose Assembly.- All hose shall be marked with a green stripe, 1/8 inch in width, and shall conform to ANA Bulletin No. 298. Dashes of the code marking shall be 1 inch in length, and spacing between dashes and dots shall be 1/4 inch. In addition to this code marking, the symbol "HP," the quarter of the year, and the year of manufacture shall be marked on the hose at intervals of not more than 12 inches.

3.7.3 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of hose assemblies shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests performed on samples submitted for approval as qualified products.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 Test conditions.-

4.2.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 70°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3 Qualification tests.-

4.3.1 Sampling instructions.- Qualification test samples shall consist of 3 inner tubes and 3 hose assemblies. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from that activity. (See paragraph 6.3.)

4.3.2 Tests.- The Qualification tests shall consist of the Acceptance tests and, in addition, the following tests.

4.3.2.1 Low temperature.- Each sample hose assembly shall be subjected to a temperature of -65°F for a period of 48 hours. While still at this temperature, the hose shall be grasped by each end and bent 180 degrees over a mandrel having a diameter of 5 inches. Any evidence of cracking shall be cause for rejection.

4.3.2.2 High temperature.- Each sample hose assembly shall be subjected to a temperature of +160°F for a period of 48 hours. After exposure to this temperature, each sample assembly shall be capable of successfully meeting the test of paragraph 4.4.1.2.

4.4 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests.- Each hose assembly shall be subjected to the following tests.

4.4.1.1 Examination of product.- Each hose assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.1.2 Pressure test.- Each hose assembly shall be subjected to a test pressure of 2,500 psi after assembly of the fittings. Pressure shall be applied for a minimum of 3 and not more than 10 minutes, with the hose assembly submerged in water. The gas used for this test shall be oxygen in accordance with Specification MIL-O-8069, oil-free dry nitrogen in accordance with Specification MIL-N-6011, or oil-free dry air. After the immediate expulsion of entrapped air from the reinforcing material, there shall be no evidence of leakage as shown by escaping gas bubbles from any part of the hose assembly. After this test, the hose assemblies shall be cleaned and dried.

4.4.2 Sampling tests.-

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4.4.2.1 Sampling plan A.-

4.4.2.1.1 Hose assemblies.- Six hose assemblies shall be selected at random from every 5,000 or fraction thereof produced and subjected to the following tests. The final test for 3 of the assemblies shall be the test of paragraph 4.4.2.1.1.1, and the final test for the remaining 3 shall be the test of paragraph 4.4.2.1.1.2.

4.4.2.1.1.1 Hose assembly tensile strength.- The complete hose assembly shall be attached by the fittings to the heads of a tensile testing machine and pulled at the rate of approximately 1 inch per minute. Any failure below a tensile pull of 1,000 pounds shall be cause for rejection.

4.4.2.1.1.2 Burst pressure.- The Burst pressure test, with the exception of the rate of pressure increase, shall be made in accordance with the hydrostatic pressure test as specified in Specification ZZ-R-601. Liquid pressure shall be applied at the rate of approximately 25,000 ±10,000 psi per minute. Any failure below a value of 15,000 psi shall be cause for rejection.

4.4.2.1.2 Inner tube sampling.- A sufficient quantity of hose shall be selected at random from each lot of 10,000 feet, or less, and shall be used to fabricate 6 dumbbells conforming to figure 2, size C, of Specification ZZ-R-601. The dumbbells shall be subjected to the tests of paragraph 4.4.2.1.2.1.

4.4.2.1.2.1 Inner tube tensile strength and elongation test.- The tensile strength and ultimate elongation of three inner tube samples shall be determined in accordance with Specification ZZ-R-601. Failure to conform to the values specified in paragraphs 3.3.2.1 and 3.3.2.2 herein shall be cause for rejection.

4.4.2.1.2.2 Accelerated aging and testing.- Three of the inner tube samples shall be aged by placing them in an oxygen bomb at a temperature of +160°F and at a pressure of 300 psi for a period of 96 hours. At the end of this period, the samples shall be removed and tested for tensile strength and ultimate elongation as specified in paragraph 4.4.2.1.2.1 herein.

4.4.2.2 Sampling plan B.-

4.4.2.2.1 Five hose assemblies shall be selected from each 100 or fraction thereof produced and subjected to the following test.

4.4.2.2.1.1 Odor tests.- A check test setup shall be fabricated as follows: A length of high-pressure copper tubing shall be used in lieu of the hose assembly for the check test setup. The copper tubing shall be connected to an oxygen cylinder charged to 2,000 psi. A type C-1 oxygen regulator conforming to Specification MIL-R-6125 to regulate the cylinder pressure to 450 psi shall be incorporated in the line. Downstream of the type C-1 regulator, a type D-2 breathing oxygen regulator conforming to Specification MIL-R-8202 shall be installed in the line. This regulator reduces 450-psi oxygen pressure to breathing pressure. An oxygen mask shall be attached to the system. The system shall then be subjected to the cylinder pressure, and the Inspector shall put on the oxygen mask and breathe the odor-free oxygen for comparison purposes when actually testing the hose assembly. After this preliminary comparison test, the copper tubing shall be replaced by the hose assembly and the gas tested for any perceptible objectionable odor. This test shall be conducted by at least three persons to obtain accurate results. Any evidence of objectionable odors shall be cause for rejection of the lot.

4.4.2.2.1.2 Dimension test.- Four 14-inch lengths of hose without attachments shall be selected at random from each lot. The inner and outer hose diameters shall be checked to determine conformance with the values specified herein. (See paragraph 3.5.)

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4.4.3 Rejection and retest.- When Sampling tests are specified on a number of items that are selected from a production run and one or more of this number fails to meet the specified tests, acceptance of all items still on hand and subsequent production will be withheld until the extent and cause of failure is determined. For operational reasons, Individual tests may be continued pending investigation of a Sampling test failure. However, final acceptance of the items on hand and subsequent production is contingent upon the Inspector's decision regarding the over-all conformance of the product to specification requirements. When corrective action has been accomplished, all necessary tests shall be repeated. If investigation indicates that defects may exist in items previously accepted, full particulars concerning the defects, including recommendations for correction, shall be furnished to the contracting officer.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging.- Each assembly shall be wrapped in commercial-grade paper. Ten wrapped assemblies shall be packed in a fiberboard container in accordance with Specification JAN-P-108. Waterproof tape in accordance with Specification JAN-P-127 shall be used to seal all open flaps and edges. Hose assemblies less than 48 inches long shall be packed straight; assemblies 48 inches long and over shall be folded into lengths of not less than 2 feet, provided the radius of each bend is not less than 3 inches. Hose assemblies of the same length only shall be packed in the same container.

5.3 Packing.- The exterior shipping containers, insofar as possible, shall contain identical items and like quantities, shall be of uniform size, and shall be designed to enclose the contents in a snug tight-fitting manner. The gross weight of the exterior shipping container, when packed for shipment, shall not exceed 200 pounds.

5.3.1 Domestic shipment.- For domestic shipment, the interior packages shall be packed in an exterior container conforming to Specification NN-B-621, NN-B-631, LLL-B-631, LLL-B-636, or PPP-B-601. When fiberboard exterior containers are used, such containers shall be fabricated from fiberboard having a Mullen test of 275 pounds or more.

5.3.2 Overseas shipment.- Unless packaging for domestic shipment is specified by the procuring activity, the interior packages shall be packed for overseas shipment in a cleated plywood box conforming to Specification JAN-P-106. Plywood, when used, shall be in accordance with Specification NN-P-515, type II, class 1. The interior of each box shall be lined with a waterproof case liner which shall be constructed of a barrier material conforming to Specification JAN-P-125. The liner shall be so constructed that its seams are equal in water repellency to that of the barrier material. The case liner shall be sealed with adhesive conforming to Specification MIL-A-140.

5.4 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Hose Assembly, High Pressure, Breathing Oxygen, Specification MIL-H-4722A, Manufacturer's Part No. *__, Federal Stock No. *__, (if no FSN available, leave space therefor).

*Information to be entered by the contractor.

5.4.1 Reinspection dates.- Reinspection marking shall be added in accordance with ANA Bulletin No. 405.

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6. NOTES

6.1 Intended use.- The hose assembly covered by this specification is intended for use as flexible connections for oxygen ground servicing equipment. The hose assembly covered by Drawing 44B27648 is intended for use in connecting units of the servicing equipment, and the hose covered by Drawing 48B3659 is for use at the servicing end of the equipment.

6.2 Ordering data.- The invitation for bids, contracts, or purchase orders should state whether the hose will be in accordance with Drawing 44B27648, 48B3659, or 54A96.

6.3 Provisions for Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on a Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to the Commander, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio the activity responsible for qualification, with a copy to the Bureau of Aeronautics, Navy Department, Washington 25, D. C.

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